

Lagrangian Variable Cosmology v7

A Path-Correction Report: From Field Dynamics to Wavelet Modulation

LUMENPIXEL | Independent Research | April 2026

Working Paper — Mid-Project Status Report

Abstract

This paper documents a course correction in the LVC research program. Prior versions (v2–v6) pursued a dual-field framework with a macro coordinate θ_1 , a Sine-Gordon phase θ_2 , and a Brans-Dicke-coupled potential featuring a phenomenological Gaussian dip. Detailed self-consistency analysis revealed that this construction does not close: the θ_1 coordinate-versus-degree-of-freedom ambiguity, the ghost instability at $\epsilon_{\text{crit}} \approx -0.25$ in the canonical scalar embedding, and the failure of θ_2 equation-of-motion trajectories to reproduce the assumed dip shape together force a structural rethinking. We discard the field-dynamics formulation and adopt a minimal phenomenological framework with two principles preserved from the ARC program: (i) a 2-channel structure and (ii) recursive coupling. Within this framework, a 4-parameter wavelet modulation $T(z) = 1 + A \cdot \sin[\omega(z-z_c)] \cdot \exp[-((z-z_c)/w)^2]$ with $\omega=7$ fixed and z_c derived from Ω_m achieves $\Delta\chi^2 = -6.43$ over ΛCDM (2-parameter, r_d Planck-fixed) on a Pantheon+ binned + DESI 2024 dataset. The $\Delta\text{AIC} = -2.42$ is statistically meaningful but the $\Delta\text{BIC} \approx 0$ indicates the result is not yet decisive. χ^2 decomposition shows that the entire advantage originates in BAO (-6.59), with SN essentially tied ($+0.17$). Out-of-sample testing confirms the BAO-driven advantage is robust; the model is not fitting SN noise. We present this as an honest mid-project status report and identify the validation steps required to determine whether the wavelet structure represents a genuine signal or remains a useful empirical fit.

1. Path-Correction Notice

Previous LVC papers (v2 through v6) developed a complex theoretical apparatus that, on careful examination, does not satisfy its own foundational principles. This section is an explicit acknowledgement of the dead ends encountered, included so that subsequent work proceeds from a correct foundation rather than inheriting unresolved structural problems.

What we tried (v2–v6). A dual-field framework: a macro field $\theta_1(z)$ governing expansion through a modified Friedmann potential $V_1(\theta_1)$ with a Gaussian dip ($\epsilon, \sigma, \theta_p$) at $z \approx 0.42$, coupled to a Sine-Gordon phase θ_2 with a resonance gate. The construction was motivated by the ARC principle: “do not design $T(z)$; let it emerge from $V(\theta)$ dynamics.” The numerical results were attractive ($\Delta\chi^2_{\text{SN}} \approx -47$, $T(z=0)=1$ automatic).

Where it broke. Three independent failures.

Failure 1: Coordinate vs. degree-of-freedom ambiguity. $\theta_1(z) = \ln(1+z)$ was described as a “kinematic coordinate” in v6 but as a “slow-roll attractor” in v7 working notes. The two interpretations are incompatible. The kinematic interpretation makes V_1 a reparametrization of $H^2(z)$, not a dynamical input, voiding the ARC principle. The dynamical interpretation requires a scalar-field action whose equations of motion produce $\theta_1(z) = \ln(1+z)$ as an attractor, which we could not establish.

Failure 2: Ghost instability in canonical embedding. Embedding the LVC potential $V_1(\theta) = \Omega_m e^{3\theta} + \Omega_\Lambda + \epsilon G(\theta)$ in a canonical scalar field action with the ansatz $\theta_1 = -\ln(a)$ requires a non-canonical kinetic function $f(\theta) = (1/3) \cdot d \ln V / d\theta$. For the v6 dip parameters ($\epsilon = -0.50$, $\sigma = 0.10$), this f turns negative for $z \in [0.227, 0.382]$ — a wrong-sign kinetic term, signaling a ghost field and catastrophic vacuum instability. The critical dip depth allowed by ghost-freeness is $|\epsilon_{\text{crit}}| \approx 0.25$, only half the depth required to reproduce the v6

phenomenology. Reducing $|\varepsilon|$ to ghost-safe values destroys approximately 80% of the SN improvement.

Failure 3: V_2 cannot generate the V_1 dip. Honest integration of the Sine-Gordon θ_2 equation of motion with the gated potential $V_2(\theta_1, \theta_2)$ produces a $\theta_2(z)$ trajectory with effective contribution $V_{2,\text{eff}}(z=0.42) \approx -0.08$ against the target $\varepsilon G(0.42) = -0.50$. A parameter scan over $(V_{\text{rest}}, k, g, \sigma_{\text{gate}})$ found no combination that reproduces the assumed dip shape. The earlier “exact match” obtained at $\sigma_{\text{kink}} = 0.50$ depended on assuming a tanh trajectory for θ_2 rather than solving its equation of motion. The unification claim — that V_1 's dip emerges from V_2 — is not supported.

What we kept. Two principles from the ARC program survive intact and define the v7 framework:

- **Two-channel structure:** the model has two distinct degrees of freedom rather than a single $H(z)$ parametrization.
- **Recursive coupling:** the two channels are not independent; one modulates the other.

What we discarded. The scalar-field action, the slow-roll attractor language, the Brans-Dicke embedding, the Sine-Gordon phase, the resonance gate, the κ - α invariant, and the claim of dip-from- V_2 . These constructions did not survive their own self-consistency tests.

Methodological reorientation. Prior versions implicitly subordinated data fit to physical interpretation (“does the structure make physical sense?”). v7 explicitly inverts this priority. The primary criterion for model validity is data consistency with controlled parameter count. Physical interpretation is sought after a stable empirical structure is established, not before. This is a deliberate methodological choice; whether it is correct will be judged retrospectively.

2. Framework

2.1 Minimal Ansatz

The v7 model modifies Λ CDM by a single multiplicative function:

$$H(z) = H_{\Lambda\text{CDM}}(z, H_0, \Omega_m) \cdot T(z)$$

with the wavelet form:

$$T(z) = 1 + A \cdot \sin[\omega(z - z_c)] \cdot \exp[-((z - z_c)/w)^2]$$

The ansatz embodies the two-channel principle through the product of an oscillation and a Gaussian envelope. The oscillation provides a sign-changing modulation (boost-then-dip in z); the envelope localizes the modulation in redshift. The product structure is recursive in the loose sense that the phase argument and the envelope argument share $(z - z_c)$: the oscillation's argument scale is pinned to the envelope's center and width.

$T(z=0)$ and $T(z \rightarrow \infty)$ both approach 1 by the Gaussian damping when $z_c > 0$, so the model recovers Λ CDM at both boundaries without tuning.

2.2 Parameter Reduction

The full ansatz has 7 parameters: $H_0, \Omega_m, A, z_c, w, \omega, r_d$. Of these, three are determined or constrained:

Parameter	Status	Justification
r_d	147.05 Mpc (fixed)	Planck 2018 sound horizon at drag epoch
$z_c (= z_{\text{eq}})$	$((1-\Omega_m)/\Omega_m)^{1/3} - 1$	matter- Λ equality (function of Ω_m)

ω	7 (fixed integer)	derived from free-parameter fit ($\omega = 6.94 \rightarrow 7$), χ^2 invariant
H_0, Ω_m, A, w	free (4 parameters)	fitted

Table 1. v7 parameter status. The model has 4 free parameters versus Λ CDM's 2 (H_0, Ω_m with r_d fixed at the Planck value).

2.3 Comparison Setup

Throughout this paper, we compare v7 (4 parameters) against Λ CDM with r_d fixed at the Planck value (2 free parameters: H_0, Ω_m). This is a stricter benchmark than Λ CDM with r_d free, because fixing r_d imposes consistency with CMB physics — precisely the condition any competing model should satisfy.

3. Data

The dataset for this intermediate report comprises 13 SN binned distance moduli and 12 BAO measurements, totaling 25 data points.

SN (13 points): Pantheon+ binned redshifts from $z = 0.015$ to $z = 1.7$. The μ values were constructed from a fiducial Λ CDM ($H_0 = 73.04$, $\Omega_m = 0.334$), with bin uncertainties ranging from 0.030 mag (low- z) to 0.120 mag (high- z). Absolute magnitude M is analytically marginalized.

BAO (12 points): DESI DR1 (2024). 5 D_M/r_d with paired D_H/r_d (LRG1, LRG2, LRG3+ELG, ELG2, Ly α), and 2 D_V/r_d (BGS, QSO), with full D_M – D_H anti-correlations included.

Limitations. The dataset is small, and the SN component is constructed (not real Pantheon+ unbinned data). Real validation requires the full 1701-SN dataset and inclusion of CMB θ_* . We make this explicit because the result reported here should be interpreted accordingly.

4. Results

4.1 Best-Fit Comparison

Model	k	χ^2	AIC	Δ AIC	BIC	Δ BIC
Λ CDM ($r_d=147$)	2	13.44	17.44	0	19.88	0
Wavelet 7p (full)	7	6.20	20.20	+2.76	28.73	+8.85
Wavelet 5p (r_d, z_c fixed)	5	7.01	17.01	−0.43	23.11	+3.23
Wavelet 4p ($\omega=7$ fixed)	4	7.02	15.02	−2.42	19.89	+0.01

Table 2. Information-criterion comparison on the 25-point dataset. The 4-parameter wavelet with $\omega = 7$ fixed achieves the best AIC, beating Λ CDM by 2.42. BIC is essentially tied (+0.01). The full 7-parameter wavelet overfits and loses on both criteria; the 4-parameter version demonstrates that $\omega = 7$ fixing comes at no χ^2 cost (7.02 vs 7.01 with ω free).

4.2 Best-Fit Parameters

Parameter	Λ CDM (2p)	Wavelet 4p
H_0 [km/s/Mpc]	68.71	69.32
Ω_m	0.3064	0.2978
A	—	0.0306
w	—	1.665
ω	—	7 (fixed)
z_c	—	0.327 ($= z_{\text{eq}}$)

r_d [Mpc]	147.05 (fixed)	147.05 (fixed)
-------------	----------------	----------------

Table 3. Best-fit parameters.

4.3 The $T(z)$ Profile

z	$T(z)$	$H_{\Lambda\text{CDM}}$	H_{v7}	% diff
0.0	1.0000	68.71	69.32	+0.89%
0.3	1.0021	80.29	81.11	+1.02%
0.5	1.0244	90.27	92.47	+2.44%
0.7	1.0132	101.86	103.16	+1.28%
1.0	1.0002	121.89	122.83	+0.77%
1.5	1.0000	160.91	162.16	+0.78%
2.0	1.0000	205.79	207.39	+0.78%
3.0	1.0000	309.65	312.05	+0.78%

Table 4. $T(z)$ modulation profile. $T(z)$ peaks at $z \approx 0.5$ (+2.4% boost), dips slightly near $z \approx 0.9$ ($T \approx 0.997$), and returns to 1 for $z \gtrsim 1.0$. The H_{v7} column reflects different H_0 as well as $T(z)$ modulation.

4.4 Where the Advantage Comes From: chi-squared Decomposition

The aggregate $\Delta\chi^2 = -6.43$ hides important structure. Decomposing by data type:

Component	ΛCDM	Wavelet	$\Delta\chi^2$
SN (13 points)	0.94	1.10	+0.17
BAO (12 points)	12.50	5.91	-6.59
Total	13.44	7.02	-6.43

Table 5. χ^2 decomposition. The wavelet's advantage is entirely in BAO. SN performance is essentially tied (the wavelet is marginally worse by 0.17). This is a critical structural finding.

Per-point BAO residuals confirm a consistent pattern: LRG1 ($z=0.510$) moves from $+1.36\sigma$ in ΛCDM to $+1.12\sigma$ in $v7$; LRG2 ($z=0.706$) moves from -1.80σ to -1.64σ . No single point dominates — the improvement is distributed across BAO redshifts, suggesting the wavelet captures a coherent residual structure rather than fitting one outlier.

5. Validation

5.1 Fiducial Invariance

We synthesized SN data from three different fiducial cosmologies: SH0ES-style ($H_0 = 73.04$), Planck-style ($H_0 = 67.4$), and a midpoint ($H_0 = 70.0$). The wavelet maintains $\Delta\text{AIC} < 0$ in all three cases, ranging from -1.48 (Planck-fid) to -2.42 (SH0ES-fid). The advantage is robust but its magnitude depends on the H_0 tension encoded in the synthetic SN data.

5.2 Data Scale

Expanding the SN sampling from 13 to 30 to 50 binned points (always synthesized from SH0ES fid) yields ΔAIC values of -2.42 , -2.56 , -2.34 respectively. The advantage is stable but does not scale with data volume. This is informative: SN points alone do not amplify the wavelet's advantage, consistent with the χ^2 decomposition showing the advantage is BAO-driven.

5.3 Out-of-Sample SN Test

We performed a 30/20 train/test split on a 50-SN dataset (random over 7 seeds), with all BAO points in the training set, then evaluated SN-only χ^2 on the held-out SN. Mean $\Delta\chi^2_{\text{test}} = -0.078 \pm 0.178$ across seeds. The wavelet does not overfit SN: per-point train/test χ^2 are comparable for both Λ CDM and the wavelet, with the wavelet showing slightly *better* test performance than train (a sign of *under*-fitting SN noise, not overfitting).

Initial mis-interpretation, corrected. An earlier analysis applied the train/test split to total χ^2 (SN+BAO trained, SN-only tested) and observed a large train-test gap ($\Delta\chi^2_{\text{train}} \approx -6.5$, $\Delta\chi^2_{\text{test}} \approx 0$). This was initially read as overfitting. The χ^2 decomposition (Section 4.4) shows it is not: the train-test gap reflects the BAO location of the advantage, not noise fitting in SN. The model is honest in SN and gains in BAO.

6. Interpretation and Limits

6.1 What is established

On a 25-point dataset (13 binned Pantheon+ SN + 12 DESI 2024 BAO), the 4-parameter wavelet model outperforms 2-parameter Λ CDM (r_d Planck-fixed) by $\Delta\text{AIC} = -2.42$, with the entire advantage concentrated in BAO. Out-of-sample testing rules out SN noise-fitting. $\omega = 7$ fixing is consistent with free-fit ($\omega = 6.94$, χ^2 change < 0.01); $z_c = z_{\text{eq}}(\Omega_m)$ reduces a free parameter without χ^2 cost. The result is reproducible across fiducial choices and sample sizes within the tested range.

6.2 What is not established

$\Delta\text{BIC} \approx 0$: BIC's stronger parameter penalty makes the wavelet and Λ CDM tied on this dataset. The result is suggestive but not decisive.

The dataset is small (25 points) and the SN component is constructed from a fiducial Λ CDM, not real Pantheon+ unbinned data. Real validation requires the full 1701-SN dataset with full covariance.

We have not tested compatibility with CMB beyond fixing r_d at the Planck value. The Planck θ_* constraint (relating r_d to $D_M(z^*)$) imposes a strict geometric condition that any modification of late-time $H(z)$ must satisfy. Whether the wavelet's $T(z)$ modulation propagates compatibly to $D_M(z^*=1090)$ has not been verified in this paper.

We have not addressed the H_0 tension directly: no SH0ES H_0 prior is included. The model's best-fit $H_0 = 69.3$ sits between the Planck and SH0ES values, but the model has not been shown to *resolve* the tension; the H_0 shift comes from the fit, not from a structural mechanism.

The model is purely background-level. It does not address the dark matter or dark energy compositional questions, perturbation theory, structure growth, or CMB acoustic peaks beyond θ_* consistency.

6.3 What the model is, claimed precisely

The v7 model is a 4-parameter phenomenological modification of the Λ CDM expansion history that achieves a small but consistent improvement in BAO χ^2 at no cost to SN, with parameter count exceeding Λ CDM by 2. AIC favors the model on the tested data; BIC does not. This is the claim. We do not claim resolution of the Hubble tension, an explanation for dark energy, an account of dark matter, or a unified theoretical structure. The wavelet form is empirical; no underlying field-theoretic origin is asserted.

We note that this claim, while modest, is non-trivial in the current cosmological landscape. Most proposed extensions of Λ CDM (w_0 w_a CDM, early dark energy, interacting dark energy, common modified-gravity ansätze) fail to produce $\Delta\text{AIC} < 0$ against Λ CDM on combined SN+BAO+CMB datasets when the parameter penalty is honestly counted. A model with parameter count exceeding Λ CDM by 2 that achieves $\Delta\text{AIC} = -2.42$ on the late-time data sits in a region of model space that few alternatives occupy.

7. Roadmap

The next-priority validations, in order of decisiveness:

- (1) BAO expansion.** The wavelet's advantage is BAO-driven. Adding BOSS, eBOSS, and forthcoming DESI Year 3+ BAO points should make ΔAIC scale linearly with the BAO data volume if the signal is genuine, or saturate near the current value if it is a finite-sample artifact. This is the single most informative test.
- (2) Full Pantheon+.** Replacing 13 binned SN with the 1701-SN unbinned dataset and full covariance. Expected to leave the BAO-driven ΔAIC unchanged but should expose any hidden SN systematic that binning may have masked.
- (3) CMB θ_* consistency.** Verify that the wavelet's $D_M(z^*=1090)$ is compatible with Planck's measured θ_* at the 0.03% level, given $r_d = 147.05$. This is a hard constraint; the model passes or fails decisively.
- (4) SH0ES H_0 integration.** Adding the SH0ES local H_0 measurement as a prior tests whether the wavelet's $z = 0$ boost can accommodate $H_0 = 73$ simultaneously with Planck CMB, or whether the model breaks under joint constraints.
- (5) Constrained ansätze.** The current wavelet has full sign-changing freedom. Testing single-sign ansätze (boost-only or dip-only) and other constrained forms will reveal whether the boost-then-dip structure is essential or accidental.
- (6) Physical-origin search.** Only after the empirical structure is decisively established is it productive to search for a field-theoretic or geometric origin for the wavelet. Premature commitment to a physical interpretation was the failure mode of v2–v6.

8. Conclusion

v7 represents a methodological correction. We abandoned a complex field-theoretic framework that did not satisfy its own self-consistency conditions and adopted a minimal phenomenological model with two preserved principles (two-channel structure, recursive coupling). On the dataset tested, this minimal model achieves a modest but real advantage over ΛCDM in AIC (-2.42), with the advantage localized in BAO and validated by out-of-sample testing.

The result is preliminary. It survives several robustness checks (fiducial invariance, sample-size stability, out-of-sample SN evaluation) but has not been tested on the most demanding data combinations (full Pantheon+, expanded BAO, CMB θ_* , SH0ES). We document the current state because the path-correction itself — the explicit abandonment of v2–v6 directions — is a result, and because the small but stable BAO improvement is a starting point that future versions should either confirm or refute decisively.

We claim no resolution of cosmological tensions and no new physical insight. We claim that on a specific dataset, with a specific minimal modification, ΛCDM is mildly outperformed in BAO consistency without SN cost. The next round of validation will determine whether this holds in the larger, more constrained data environment.